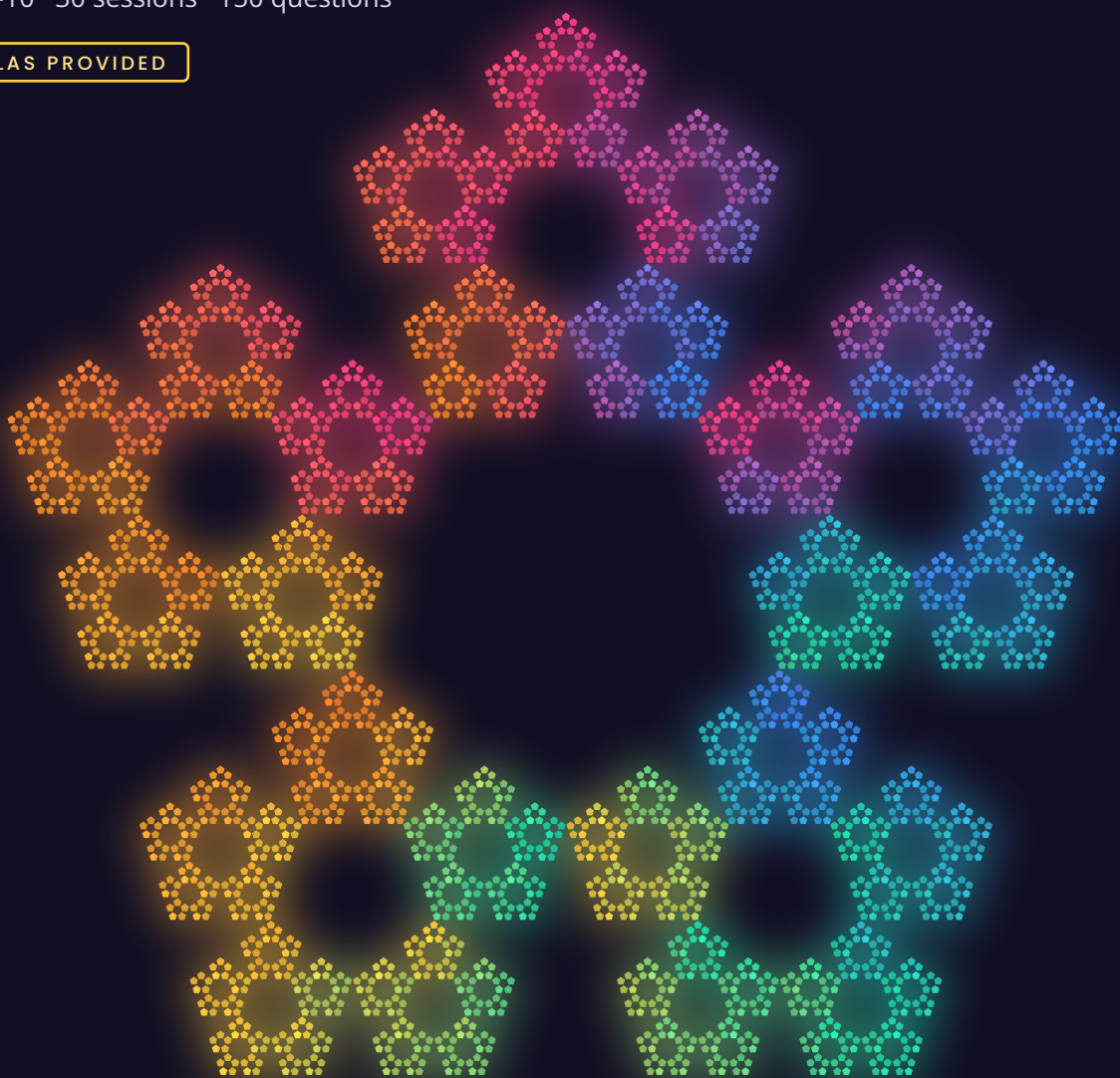


FIVE QUESTIONS. EVERY LESSON.

The Fractal Five

Weeks 1–10 · 30 sessions · 150 questions

FORMULAS PROVIDED



NAME _____

CLASS _____

SIERPIŃSKI PENTAGON

five copies of itself, each scaled by $1/\varphi^2 \approx 0.382$

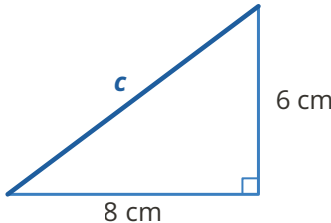
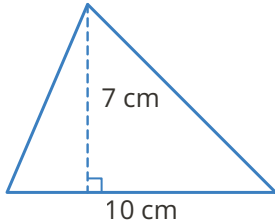
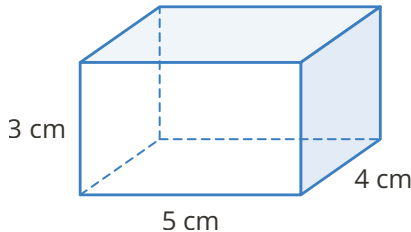
Week 1 · Session 1

TERM 1

Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING						
1 Find the length of the hypotenuse, c. 	FORMULA $c^2 = a^2 + b^2$ COPY SUBSTITUTE ANSWER						
2 Find the area of this triangle. 	FORMULA $A = \frac{1}{2}bh$ COPY SUBSTITUTE ANSWER						
3 Find the volume of this rectangular prism. 	FORMULA $V = lwh$ COPY SUBSTITUTE ANSWER						
4 Calculate. 15% of \$80	FORMULA $\text{Amount} = \frac{\text{percentage}}{100} \times \text{quantity}$ COPY SUBSTITUTE ANSWER						
5 Find the mean of these scores. <table><tr><td>Scores</td><td>4</td><td>7</td><td>9</td><td>6</td><td>9</td></tr></table>	Scores	4	7	9	6	9	FORMULA $\text{Mean} = \frac{\text{sum of scores}}{\text{number of scores}}$ COPY SUBSTITUTE ANSWER
Scores	4	7	9	6	9		

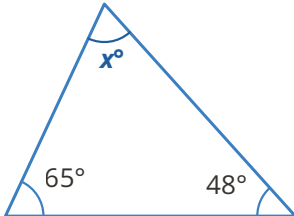
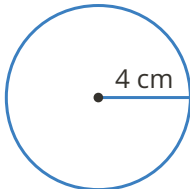
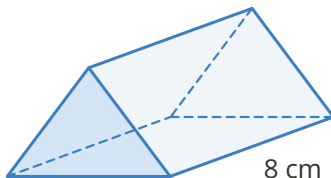
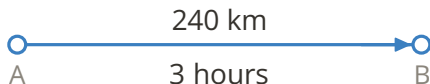
Week 1 · Session 2

TERM 1

Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING								
1 Find the value of x. 	FORMULA $a + b + c = 180^\circ$ COPY SUBSTITUTE ANSWER								
2 Find the area of this circle, correct to 1 decimal place. 	FORMULA $A = \pi r^2$ COPY SUBSTITUTE ANSWER								
3 The shaded face has an area of 12 cm^2. Find the volume of the prism. 	FORMULA $V = Ah$ COPY SUBSTITUTE ANSWER								
4 A car travels from A to B. Find its average speed. 	FORMULA $\text{Speed} = \frac{\text{distance}}{\text{time}}$ COPY SUBSTITUTE ANSWER								
5 One marble is chosen at random from this bag. Find the probability that it is blue. <table border="1" data-bbox="229 1910 679 2033"><tr><td>Colour</td><td>Red</td><td>Blue</td><td>Green</td></tr><tr><td>Marbles</td><td>3</td><td>5</td><td>2</td></tr></table>	Colour	Red	Blue	Green	Marbles	3	5	2	FORMULA $P(E) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$ COPY SUBSTITUTE ANSWER
Colour	Red	Blue	Green						
Marbles	3	5	2						

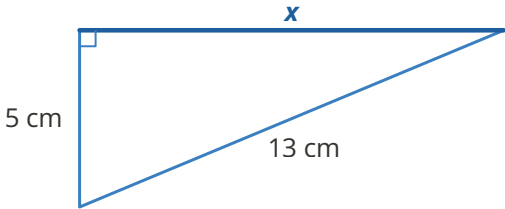
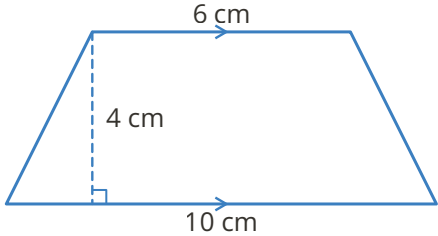
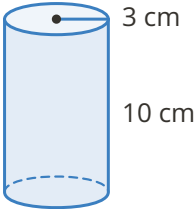
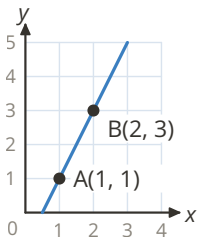
Week 1 · Session 3

TERM 1

Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA $a^2 = c^2 - b^2$ COPY _____ SUBSTITUTE _____ ANSWER _____
2 Find the area of this trapezium. 	FORMULA $A = \frac{1}{2}h(a + b)$ COPY _____ SUBSTITUTE _____ ANSWER _____
3 Find the volume of this cylinder, correct to 1 decimal place. 	FORMULA $V = \pi r^2 h$ COPY _____ SUBSTITUTE _____ ANSWER _____
4 Write as a single power. $5^4 \times 5^3$	FORMULA $a^m \times a^n = a^{m+n}$ COPY _____ SUBSTITUTE _____ ANSWER _____
5 Find the gradient of line AB. 	FORMULA $m = \frac{\text{rise}}{\text{run}}$ COPY _____ SUBSTITUTE _____ ANSWER _____

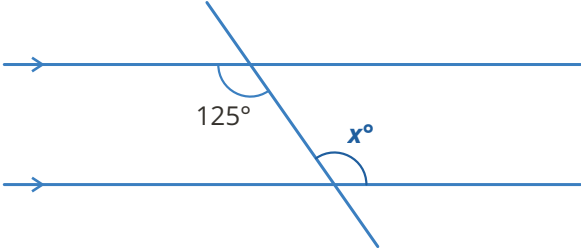
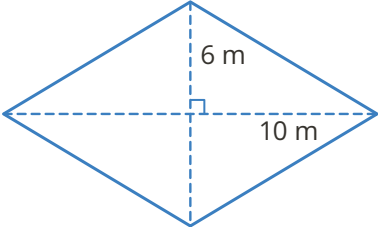
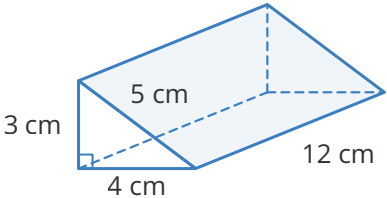
Week 2 · Session 1

TERM 1

Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA Alternate angles on parallel lines are equal COPY _____ SUBSTITUTE _____ ANSWER _____
2 Find the area of this rhombus. 	FORMULA $A = \frac{1}{2}xy$ COPY _____ SUBSTITUTE _____ ANSWER _____
3 Find the surface area of this triangular prism. 	FORMULA $SA = 2 \times \text{area of triangle} + \text{area of the 3 rectangles}$ COPY _____ SUBSTITUTE _____ ANSWER _____
4 Expand. $6(a + 1)$	FORMULA $a(b + c) = ab + ac$ COPY _____ SUBSTITUTE _____ ANSWER _____
5 Find the value of y when $x = 1$. $y = 5x + 2$	FORMULA $y = mx + c$ COPY _____ SUBSTITUTE _____ ANSWER _____

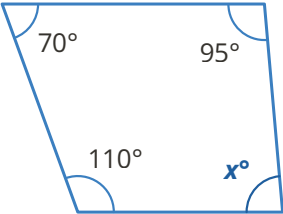
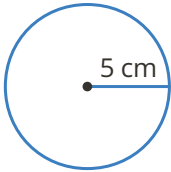
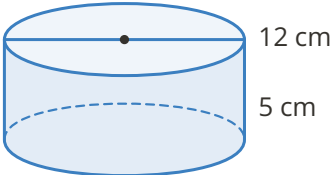
Week 2 · Session 2

TERM 1

Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING				
1 Find the value of x. 	FORMULA $a + b + c + d = 360^\circ$ COPY _____ SUBSTITUTE _____ ANSWER _____				
2 Find the circumference of this circle, correct to 1 decimal place. 	FORMULA $C = 2\pi r$ COPY _____ SUBSTITUTE _____ ANSWER _____				
3 Find the surface area of this closed cylinder, correct to 1 decimal place. 	FORMULA $SA = 2\pi r^2 + 2\pi rh$ COPY _____ SUBSTITUTE _____ ANSWER _____				
4 Find the percentage increase in price. <table border="1" data-bbox="312 1543 603 1668"> <tr> <td>Original price</td><td>\$25</td></tr> <tr> <td>New price</td><td>\$30</td></tr> </table>	Original price	\$25	New price	\$30	FORMULA $\% \text{ change} = \frac{\text{change}}{\text{original}} \times 100\%$ COPY _____ SUBSTITUTE _____ ANSWER _____
Original price	\$25				
New price	\$30				
5 Find the probability that the team does not win. $P(\text{win}) = 0.35$	FORMULA $P(\text{not } E) = 1 - P(E)$ COPY _____ SUBSTITUTE _____ ANSWER _____				

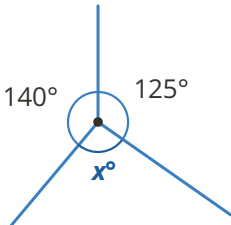
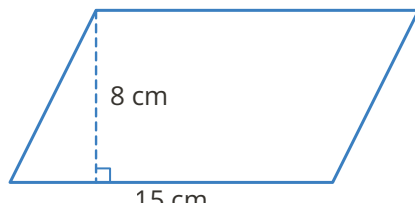
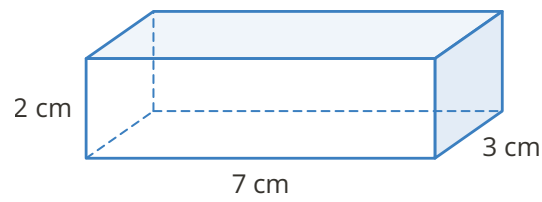
Week 2 · Session 3

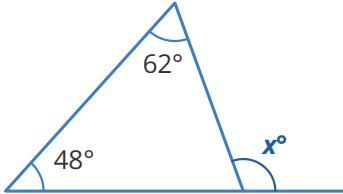
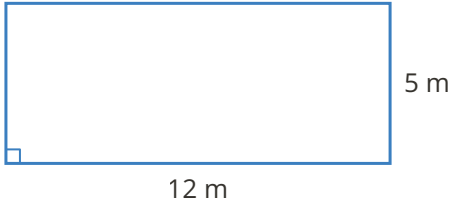
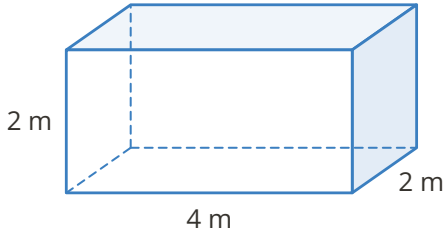
TERM 1

Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING						
1 Find the value of x. 	FORMULA Angles at a point add to 360° COPY SUBSTITUTE ANSWER						
2 Find the area of this parallelogram. 	FORMULA $A = bh$ COPY SUBSTITUTE ANSWER						
3 Find the surface area of this prism. 	FORMULA $SA = 2(lw + lh + wh)$ COPY SUBSTITUTE ANSWER						
4 Find the sale price. <table border="1" data-bbox="309 1532 595 1666"><tr><td>Original price</td><td>\$400</td></tr><tr><td>Discount</td><td>15%</td></tr></table>	Original price	\$400	Discount	15%	FORMULA New amount = original $\times$ (100% – decrease) COPY SUBSTITUTE ANSWER		
Original price	\$400						
Discount	15%						
5 Find the range of these goal tallies. <table border="1" data-bbox="150 1912 754 1980"><tr><td>Goals</td><td>5</td><td>11</td><td>8</td><td>2</td><td>9</td></tr></table>	Goals	5	11	8	2	9	FORMULA Range = highest – lowest COPY SUBSTITUTE ANSWER
Goals	5	11	8	2	9		

QUESTION	WORKING				
1 Find the value of x. 	FORMULA Exterior angle = sum of the two interior opposite angles COPY _____ SUBSTITUTE _____ ANSWER _____				
2 Find the area of this rectangle. 	FORMULA $A = lw$ COPY _____ SUBSTITUTE _____ ANSWER _____				
3 Find the capacity of this tank in litres. 	FORMULA $V = lwh$, $1 \text{ m}^3 = 1000 \text{ L}$ COPY _____ SUBSTITUTE _____ ANSWER _____				
4 Find the loss made on this sale. <table border="1" data-bbox="317 1543 595 1666"> <tr> <td>Cost price</td><td>\$95</td></tr> <tr> <td>Selling price</td><td>\$80</td></tr> </table>	Cost price	\$95	Selling price	\$80	FORMULA Profit = selling price – cost price COPY _____ SUBSTITUTE _____ ANSWER _____
Cost price	\$95				
Selling price	\$80				
5 Find the value of y when $x = 0$. $y = 3x + 4$	FORMULA $y = mx + c$ COPY _____ SUBSTITUTE _____ ANSWER _____				

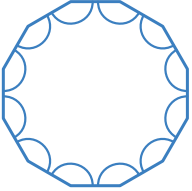
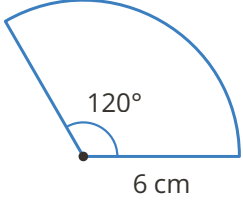
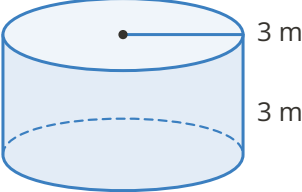
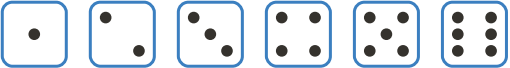
Week 3 · Session 2

TERM 1

Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING				
1 Find the sum of the interior angles of this dodecagon. 	FORMULA Angle sum = $180^\circ \times (n - 2)$ COPY _____ SUBSTITUTE _____ ANSWER _____				
2 Find the area of this sector, correct to 1 decimal place. 	FORMULA $A = \frac{\theta}{360} \times \pi r^2$ COPY _____ SUBSTITUTE _____ ANSWER _____				
3 Find the surface area of this closed cylinder, correct to 1 decimal place. 	FORMULA $SA = 2\pi r^2 + 2\pi rh$ COPY _____ SUBSTITUTE _____ ANSWER _____				
4 Find the rent before the increase. <table border="1" data-bbox="339 1545 579 1668"> <tr> <td>Increase</td><td>10%</td></tr> <tr> <td>New rent</td><td>\$440</td></tr> </table>	Increase	10%	New rent	\$440	FORMULA Original amount = known amount $\div$ its percentage $\times 100$ COPY _____ SUBSTITUTE _____ ANSWER _____
Increase	10%				
New rent	\$440				
5 A fair die is rolled. Find the probability of rolling a number less than 3. 	FORMULA $P(E) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$ COPY _____ SUBSTITUTE _____ ANSWER _____				

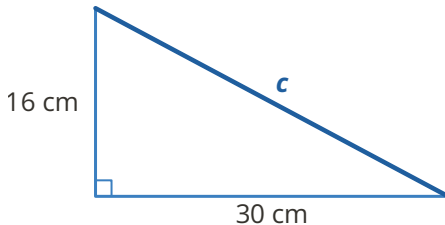
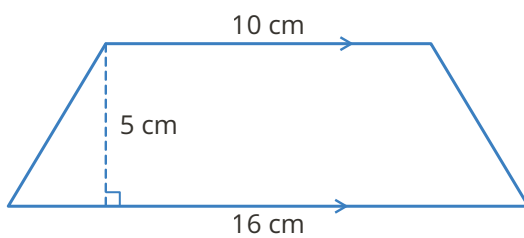
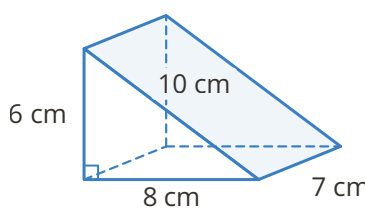
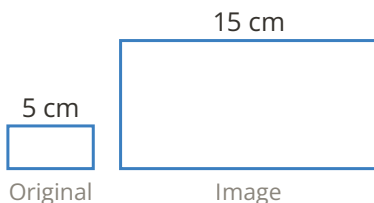
Week 3 · Session 3

TERM 1

Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING						
1 Find the length of the hypotenuse, c. 	FORMULA $c^2 = a^2 + b^2$ COPY SUBSTITUTE ANSWER						
2 Find the area of this trapezium. 	FORMULA $A = \frac{1}{2}h(a + b)$ COPY SUBSTITUTE ANSWER						
3 Find the surface area of this triangular prism. 	FORMULA $SA = 2 \times \text{area of triangle} + \text{area of the 3 rectangles}$ COPY SUBSTITUTE ANSWER						
4 The small rectangle is enlarged to make the large one. Find the scale factor. 	FORMULA $\text{Scale factor} = \frac{\text{image length}}{\text{original length}}$ COPY SUBSTITUTE ANSWER						
5 Find the range of these scores. <table><tr><td>Scores</td><td>9</td><td>3</td><td>12</td><td>7</td><td>5</td></tr></table>	Scores	9	3	12	7	5	FORMULA $\text{Range} = \text{highest} - \text{lowest}$ COPY SUBSTITUTE ANSWER
Scores	9	3	12	7	5		

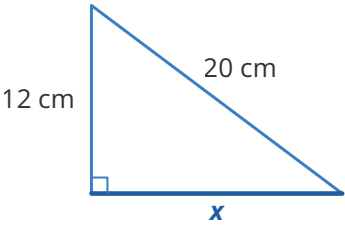
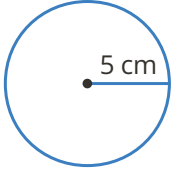
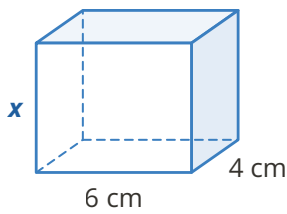
Week 4 · Session 1

TERM 1

Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA $a^2 = c^2 - b^2$ COPY _____ SUBSTITUTE _____ _____ ANSWER _____
2 Find the area of this circle, correct to 1 decimal place. 	FORMULA $A = \pi r^2$ COPY _____ SUBSTITUTE _____ _____ ANSWER _____
3 The volume of this rectangular prism is 120 cm^3. Find x. 	FORMULA $V = lwh$ COPY _____ SUBSTITUTE _____ _____ ANSWER _____
4 Calculate. 50% of 86 kg	FORMULA Amount = $\frac{\text{percentage}}{100} \times \text{quantity}$ COPY _____ SUBSTITUTE _____ _____ ANSWER _____
5 Find the probability that the bus is not late. $P(\text{late}) = 0.15$	FORMULA $P(\text{not } E) = 1 - P(E)$ COPY _____ SUBSTITUTE _____ _____ ANSWER _____

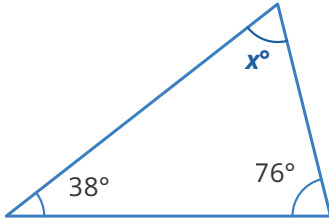
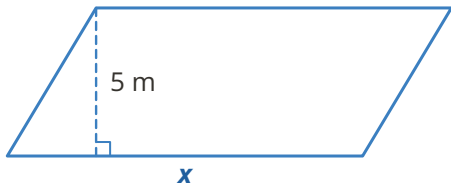
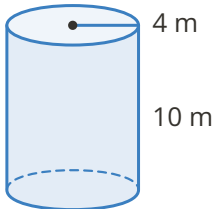
Week 4 · Session 2

TERM 1

Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING						
1 Find the value of x. 	FORMULA $a + b + c = 180^\circ$ COPY SUBSTITUTE ANSWER						
2 The area of this parallelogram is 60 m^2. Find x. 	FORMULA $A = bh$ COPY SUBSTITUTE ANSWER						
3 Find the volume of this cylinder, correct to 1 decimal place. 	FORMULA $V = \pi r^2 h$ COPY SUBSTITUTE ANSWER						
4 Evaluate. 7^0	FORMULA $a^0 = 1$ COPY SUBSTITUTE ANSWER						
5 Find the mean of these goal tallies. <table border="1" data-bbox="150 1921 761 1986"><tr><td>Goals</td><td>6</td><td>2</td><td>9</td><td>4</td><td>9</td></tr></table>	Goals	6	2	9	4	9	FORMULA Mean = $\frac{\text{sum of scores}}{\text{number of scores}}$ COPY SUBSTITUTE ANSWER
Goals	6	2	9	4	9		

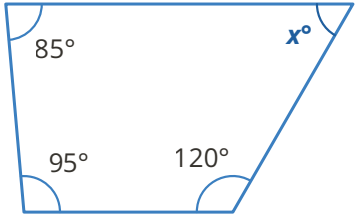
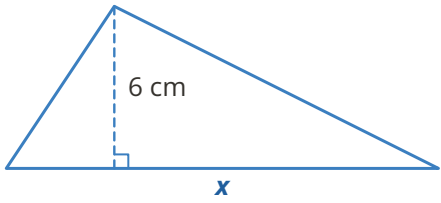
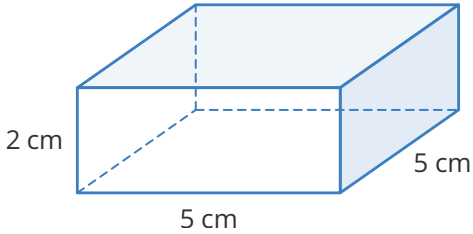
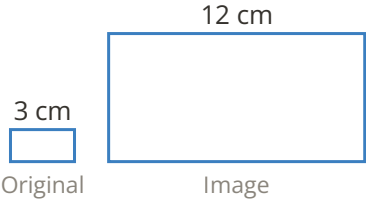
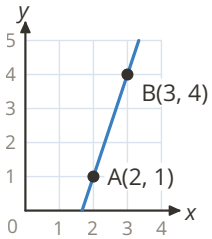
Week 4 · Session 3

TERM 1

Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA $a + b + c + d = 360^\circ$ COPY _____ SUBSTITUTE _____ ANSWER _____
2 The area of this triangle is 48 cm^2. Find x. 	FORMULA $A = \frac{1}{2}bh$ COPY _____ SUBSTITUTE _____ ANSWER _____
3 Find the surface area of this prism. 	FORMULA $SA = 2(lw + lh + wh)$ COPY _____ SUBSTITUTE _____ ANSWER _____
4 The small rectangle is enlarged to make the large one. Find the scale factor. 	FORMULA $\text{Scale factor} = \frac{\text{image length}}{\text{original length}}$ COPY _____ SUBSTITUTE _____ ANSWER _____
5 Find the gradient of line AB. 	FORMULA $m = \frac{\text{rise}}{\text{run}}$ COPY _____ SUBSTITUTE _____ ANSWER _____

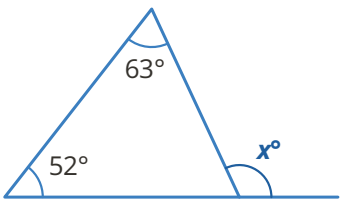
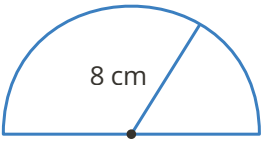
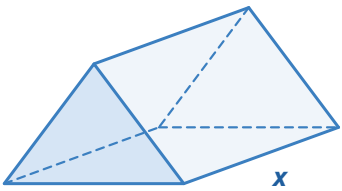
Week 5 · Session 1

TERM 1

Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING				
1 Find the value of x. 	FORMULA Exterior angle = sum of the two interior opposite angles COPY _____ SUBSTITUTE _____ ANSWER _____				
2 Find the area of this semicircle, correct to 1 decimal place. 	FORMULA $A = \frac{1}{2}\pi r^2$ COPY _____ SUBSTITUTE _____ ANSWER _____				
3 The shaded face has an area of 12 cm^2 and the volume is 96 cm^3. Find x. 	FORMULA $V = Ah$ COPY _____ SUBSTITUTE _____ ANSWER _____				
4 Find the original price of the backpack. <table border="1" data-bbox="331 1541 582 1668"> <tr> <td>Discount</td><td>30%</td></tr> <tr> <td>Sale price</td><td>\$56</td></tr> </table>	Discount	30%	Sale price	\$56	FORMULA Original amount = known amount $\div$ its percentage $\times 100$ COPY _____ SUBSTITUTE _____ ANSWER _____
Discount	30%				
Sale price	\$56				
5 Find the probability that the counter is not red. $P(\text{red}) = \frac{1}{4}$	FORMULA $P(\text{not } E) = 1 - P(E)$ COPY _____ SUBSTITUTE _____ ANSWER _____				

Week 5 · Session 2

TERM 1

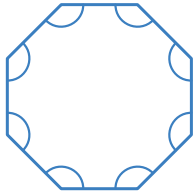
Copy the formula, substitute, then solve.

NAME _____

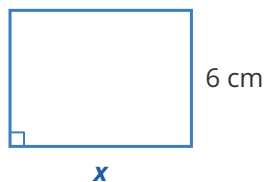
DATE _____

QUESTION

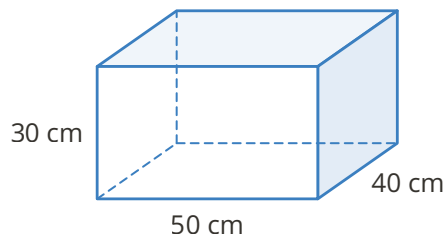
- 1 Find the sum of the interior angles of this octagon.



- 2 The area of this rectangle is 48 cm^2 . Find x .



- 3 Find the capacity of this tank in litres.



- 4 Find the profit made on this sale.

Cost price	\$18
Selling price	\$30

- 5 One marble is chosen at random from this bag. Find the probability that it is green.

Colour	Red	Blue	Green
Marbles	4	2	6

WORKING

FORMULA Angle sum = $180^\circ \times (n - 2)$

COPY _____

SUBSTITUTE _____

ANSWER _____

FORMULA $A = lw$

COPY _____

SUBSTITUTE _____

ANSWER _____

FORMULA $V = lwh$, $1000 \text{ cm}^3 = 1 \text{ L}$

COPY _____

SUBSTITUTE _____

ANSWER _____

FORMULA Profit = selling price – cost price

COPY _____

SUBSTITUTE _____

ANSWER _____

FORMULA $P(E) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$

COPY _____

SUBSTITUTE _____

ANSWER _____

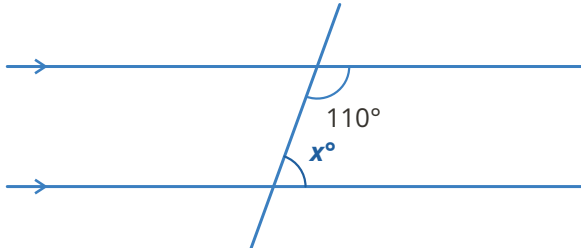
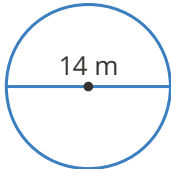
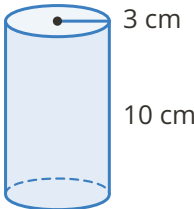
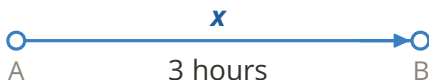
Week 5 · Session 3

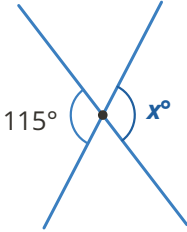
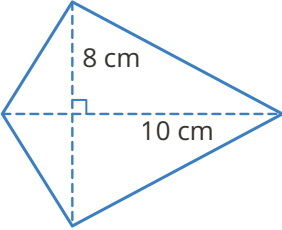
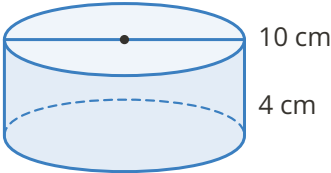
TERM 1

Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING						
1 Find the value of x. 	FORMULA Co-interior angles on parallel lines add to 180° COPY SUBSTITUTE ANSWER						
2 Find the circumference of this circle, correct to 1 decimal place. 	FORMULA $C = \pi d$ COPY SUBSTITUTE ANSWER						
3 Find the surface area of this closed cylinder, correct to 1 decimal place. 	FORMULA $SA = 2\pi r^2 + 2\pi rh$ COPY SUBSTITUTE ANSWER						
4 A car travels from A to B at an average speed of 80 km/h. Find x. 	FORMULA $\text{Speed} = \frac{\text{distance}}{\text{time}}$ COPY SUBSTITUTE ANSWER						
5 Find the range of these marks. <table><tr><td>Marks</td><td>61</td><td>47</td><td>55</td><td>70</td><td>52</td></tr></table>	Marks	61	47	55	70	52	FORMULA Range = highest – lowest COPY SUBSTITUTE ANSWER
Marks	61	47	55	70	52		

QUESTION	WORKING				
1 Find the value of x. 	FORMULA Vertically opposite angles are equal COPY _____ SUBSTITUTE _____ ANSWER _____				
2 Find the area of this kite. 	FORMULA $A = \frac{1}{2}xy$ COPY _____ SUBSTITUTE _____ ANSWER _____				
3 Find the volume of this cylinder, correct to 1 decimal place. 	FORMULA $V = \pi r^2 h$ COPY _____ SUBSTITUTE _____ ANSWER _____				
4 Find the percentage increase in visitors. <table border="1" data-bbox="331 1543 582 1668"> <tr> <td>Last year</td><td>1,200</td></tr> <tr> <td>This year</td><td>1,500</td></tr> </table>	Last year	1,200	This year	1,500	FORMULA $\% \text{ change} = \frac{\text{change}}{\text{original}} \times 100\%$ COPY _____ SUBSTITUTE _____ ANSWER _____
Last year	1,200				
This year	1,500				
5 Find the value of y when $x = 2$. $y = 6x + 1$	FORMULA $y = mx + c$ COPY _____ SUBSTITUTE _____ ANSWER _____				

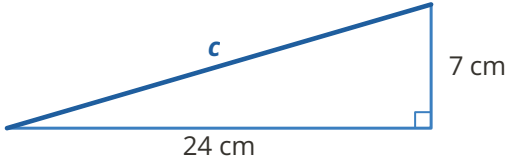
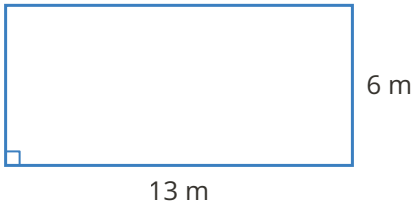
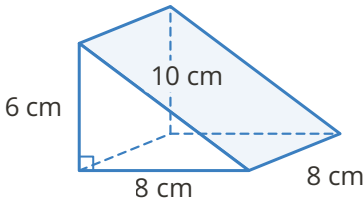
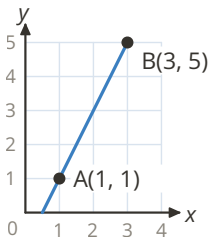
Week 6 · Session 2

TERM 1

Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the length of the hypotenuse, c. 	FORMULA $c^2 = a^2 + b^2$ COPY _____ SUBSTITUTE _____ ANSWER _____
2 Find the area of this rectangle. 	FORMULA $A = lw$ COPY _____ SUBSTITUTE _____ ANSWER _____
3 Find the surface area of this triangular prism. 	FORMULA $SA = 2 \times \text{area of triangle} + \text{area of the 3 rectangles}$ COPY _____ SUBSTITUTE _____ ANSWER _____
4 Expand. $7(y + 3)$	FORMULA $a(b + c) = ab + ac$ COPY _____ SUBSTITUTE _____ ANSWER _____
5 Find the gradient of line AB. 	FORMULA $m = \frac{\text{rise}}{\text{run}}$ COPY _____ SUBSTITUTE _____ ANSWER _____

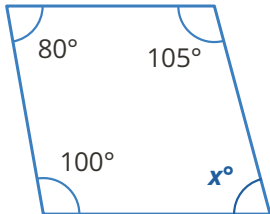
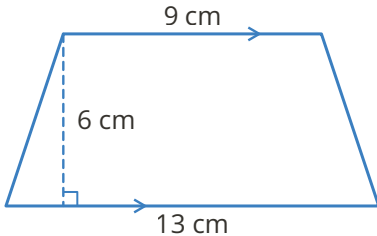
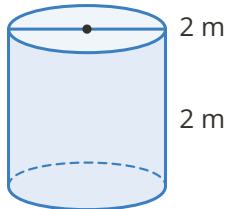
Week 6 · Session 3

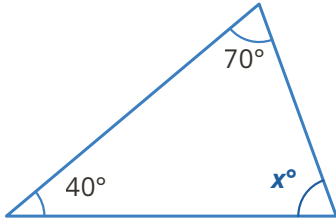
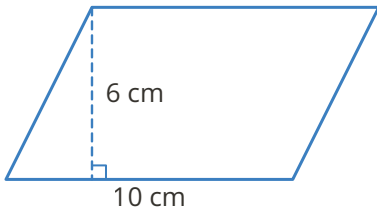
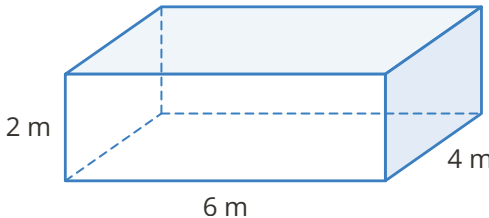
TERM 1

Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING						
1 Find the value of x. 	FORMULA $a + b + c + d = 360^\circ$ COPY SUBSTITUTE ANSWER						
2 Find the area of this trapezium. 	FORMULA $A = \frac{1}{2}h(a + b)$ COPY SUBSTITUTE ANSWER						
3 Find the capacity of this tank in litres, correct to 1 decimal place. 	FORMULA $V = \pi r^2 h, \quad 1 \text{ m}^3 = 1000 \text{ L}$ COPY SUBSTITUTE ANSWER						
4 Find the new population of the town. <table><tr><td>Population</td><td>2,400</td></tr><tr><td>Increase</td><td>5%</td></tr></table>	Population	2,400	Increase	5%	FORMULA New amount = original $\times$ (100% + increase) COPY SUBSTITUTE ANSWER		
Population	2,400						
Increase	5%						
5 Find the mean of these scores. <table><tr><td>Scores</td><td>13</td><td>19</td><td>16</td><td>10</td><td>17</td></tr></table>	Scores	13	19	16	10	17	FORMULA Mean = $\frac{\text{sum of scores}}{\text{number of scores}}$ COPY SUBSTITUTE ANSWER
Scores	13	19	16	10	17		

QUESTION	WORKING						
1 Find the value of x. 	FORMULA $a + b + c = 180^\circ$ COPY SUBSTITUTE ANSWER						
2 Find the area of this parallelogram. 	FORMULA $A = bh$ COPY SUBSTITUTE ANSWER						
3 Find the surface area of this prism. 	FORMULA $SA = 2(lw + lh + wh)$ COPY SUBSTITUTE ANSWER						
4 Calculate. 75% of \$48	FORMULA Amount = $\frac{\text{percentage}}{100} \times \text{quantity}$ COPY SUBSTITUTE ANSWER						
5 Find the range of these scores. <table><tr><td>Scores</td><td>40</td><td>38</td><td>45</td><td>33</td><td>41</td></tr></table>	Scores	40	38	45	33	41	FORMULA Range = highest – lowest COPY SUBSTITUTE ANSWER
Scores	40	38	45	33	41		

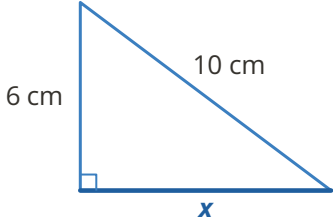
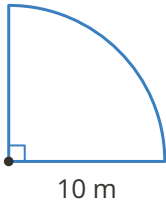
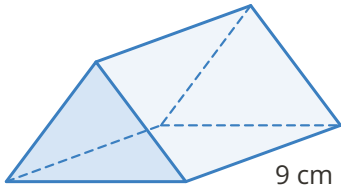
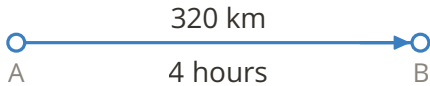
Week 7 · Session 2

TERM 1

Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA $a^2 = c^2 - b^2$ COPY _____ SUBSTITUTE _____ ANSWER _____
2 Find the area of this quadrant, correct to 1 decimal place. 	FORMULA $A = \frac{\theta}{360} \times \pi r^2$ COPY _____ SUBSTITUTE _____ ANSWER _____
3 The shaded face has an area of 12 cm^2. Find the volume of the prism. 	FORMULA $V = Ah$ COPY _____ SUBSTITUTE _____ ANSWER _____
4 A car travels from A to B. Find the average speed. 	FORMULA $\text{Speed} = \frac{\text{distance}}{\text{time}}$ COPY _____ SUBSTITUTE _____ ANSWER _____
5 Find the probability that a goal is not scored. $P(\text{goal}) = \frac{2}{5}$	FORMULA $P(\text{not } E) = 1 - P(E)$ COPY _____ SUBSTITUTE _____ ANSWER _____

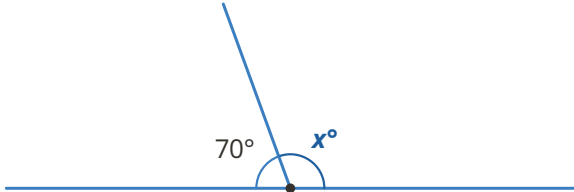
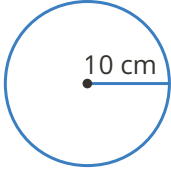
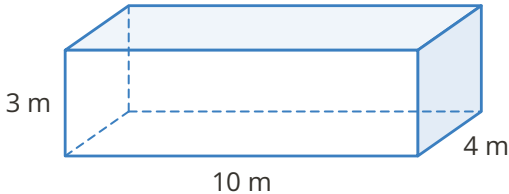
Week 7 · Session 3

TERM 1

Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA Angles on a straight line add to 180° COPY _____ SUBSTITUTE _____ ANSWER _____
2 Find the area of this circle, correct to 1 decimal place. 	FORMULA $A = \pi r^2$ COPY _____ SUBSTITUTE _____ ANSWER _____
3 Find the volume of this rectangular prism. 	FORMULA $V = lwh$ COPY _____ SUBSTITUTE _____ ANSWER _____
4 Write as a single power. $2^9 \div 2^4$	FORMULA $a^m \div a^n = a^{m-n}$ COPY _____ SUBSTITUTE _____ ANSWER _____
5 Find the value of y when $x = 2$. $y = 4x - 1$	FORMULA $y = mx + c$ COPY _____ SUBSTITUTE _____ ANSWER _____

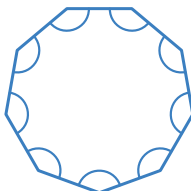
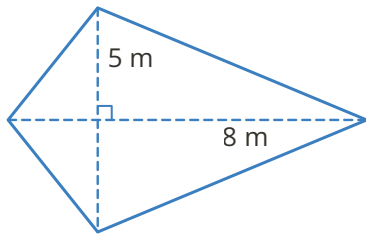
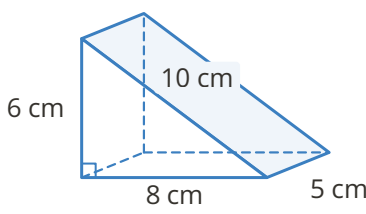
Week 8 · Session 1

TERM 1

Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the sum of the interior angles of this nonagon. 	FORMULA Angle sum = $180^\circ \times (n - 2)$ COPY _____ SUBSTITUTE _____ ANSWER _____
2 Find the area of this kite. 	FORMULA $A = \frac{1}{2}xy$ COPY _____ SUBSTITUTE _____ ANSWER _____
3 Find the surface area of this triangular prism. 	FORMULA $SA = 2 \times \text{area of triangle} + \text{area of the 3 rectangles}$ COPY _____ SUBSTITUTE _____ ANSWER _____
4 Expand. $2(3x + 5)$	FORMULA $a(b + c) = ab + ac$ COPY _____ SUBSTITUTE _____ ANSWER _____
5 A fair die is rolled. Find the probability of rolling an even number. 	FORMULA $P(E) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$ COPY _____ SUBSTITUTE _____ ANSWER _____

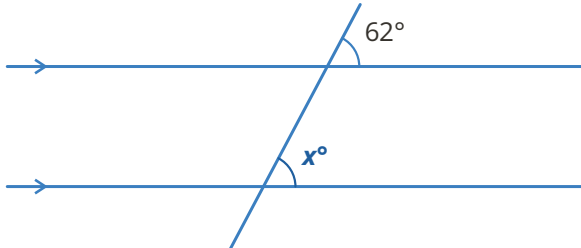
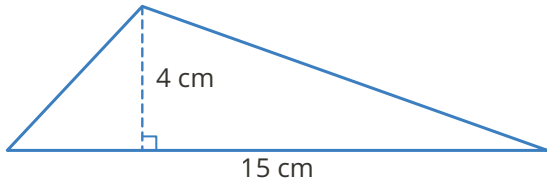
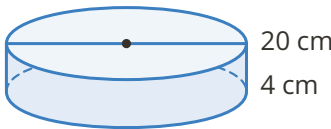
Week 8 · Session 2

TERM 1

Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING						
1 Find the value of x. 	FORMULA Corresponding angles on parallel lines are equal COPY SUBSTITUTE ANSWER						
2 Find the area of this triangle. 	FORMULA $A = \frac{1}{2}bh$ COPY SUBSTITUTE ANSWER						
3 Find the surface area of this closed cylinder, correct to 1 decimal place. 	FORMULA $SA = 2\pi r^2 + 2\pi rh$ COPY SUBSTITUTE ANSWER						
4 Find the new price. <table><tr><td>Original price</td><td>\$80</td></tr><tr><td>Increase</td><td>15%</td></tr></table>	Original price	\$80	Increase	15%	FORMULA New amount = original $\times$ (100% + increase) COPY SUBSTITUTE ANSWER		
Original price	\$80						
Increase	15%						
5 Find the mean of these marks. <table><tr><td>Marks</td><td>14</td><td>11</td><td>17</td><td>12</td><td>16</td></tr></table>	Marks	14	11	17	12	16	FORMULA Mean = $\frac{\text{sum of scores}}{\text{number of scores}}$ COPY SUBSTITUTE ANSWER
Marks	14	11	17	12	16		

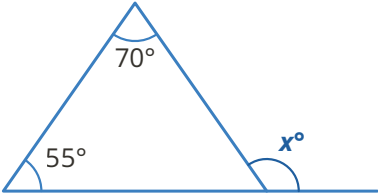
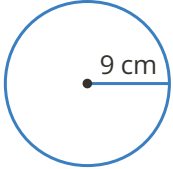
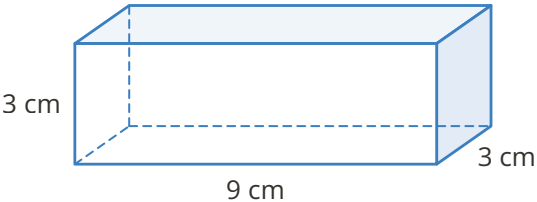
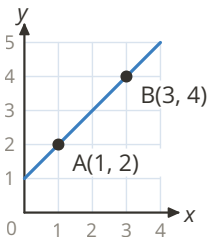
Week 8 · Session 3

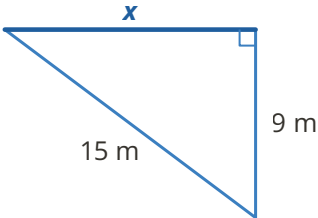
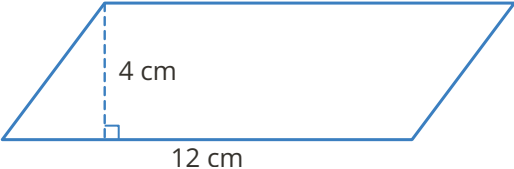
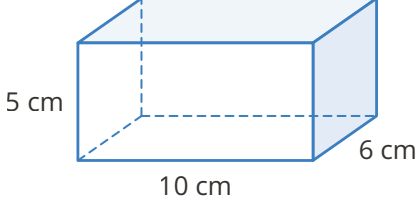
TERM 1

Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING				
1 Find the value of x. 	FORMULA Exterior angle = sum of the two interior opposite angles COPY _____ SUBSTITUTE _____ ANSWER _____				
2 Find the circumference of this circle, correct to 1 decimal place. 	FORMULA $C = 2\pi r$ COPY _____ SUBSTITUTE _____ ANSWER _____				
3 Find the surface area of this prism. 	FORMULA $SA = 2(lw + lh + wh)$ COPY _____ SUBSTITUTE _____ ANSWER _____				
4 Find the percentage decrease in price. <table border="1" data-bbox="311 1543 603 1668"> <tr> <td>Original price</td><td>\$120</td></tr> <tr> <td>New price</td><td>\$90</td></tr> </table>	Original price	\$120	New price	\$90	FORMULA $\% \text{ change} = \frac{\text{change}}{\text{original}} \times 100\%$ COPY _____ SUBSTITUTE _____ ANSWER _____
Original price	\$120				
New price	\$90				
5 Find the gradient of line AB. 	FORMULA $m = \frac{\text{rise}}{\text{run}}$ COPY _____ SUBSTITUTE _____ ANSWER _____				

QUESTION	WORKING				
1 Find the value of x. 	FORMULA $a^2 = c^2 - b^2$ COPY _____ SUBSTITUTE _____ ANSWER _____				
2 Find the area of this parallelogram. 	FORMULA $A = bh$ COPY _____ SUBSTITUTE _____ ANSWER _____				
3 Find the capacity of this container in millilitres. 	FORMULA $V = lwh$, $1 \text{ cm}^3 = 1 \text{ mL}$ COPY _____ SUBSTITUTE _____ ANSWER _____				
4 Find the profit made on this sale. <table border="1" data-bbox="316 1541 598 1668"> <tr> <td>Cost price</td><td>\$125</td></tr> <tr> <td>Selling price</td><td>\$160</td></tr> </table>	Cost price	\$125	Selling price	\$160	FORMULA Profit = selling price – cost price COPY _____ SUBSTITUTE _____ ANSWER _____
Cost price	\$125				
Selling price	\$160				
5 Find the value of y when $x = 1$. $y = 10x - 4$	FORMULA $y = mx + c$ COPY _____ SUBSTITUTE _____ ANSWER _____				

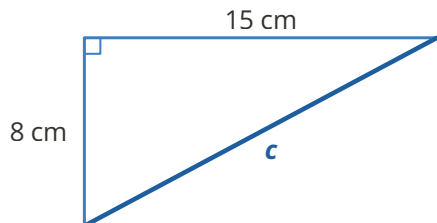
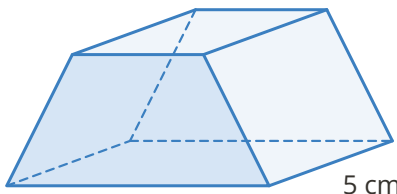
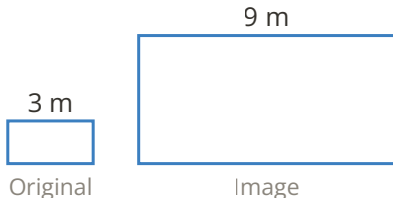
Week 9 · Session 2

TERM 1

Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING						
1 Find the length of the hypotenuse, c. 	FORMULA $c^2 = a^2 + b^2$ COPY SUBSTITUTE ANSWER						
2 Find the area of this rectangle. 	FORMULA $A = lw$ COPY SUBSTITUTE ANSWER						
3 The shaded face has an area of 24 cm^2. Find the volume of the prism. 	FORMULA $V = Ah$ COPY SUBSTITUTE ANSWER						
4 The small rectangle is enlarged to make the large one. Find the scale factor. 	FORMULA Scale factor = $\frac{\text{image length}}{\text{original length}}$ COPY SUBSTITUTE ANSWER						
5 Find the range of these heights. <table><tr><td>cm</td><td>104</td><td>96</td><td>118</td><td>101</td><td>99</td></tr></table>	cm	104	96	118	101	99	FORMULA Range = highest – lowest COPY SUBSTITUTE ANSWER
cm	104	96	118	101	99		

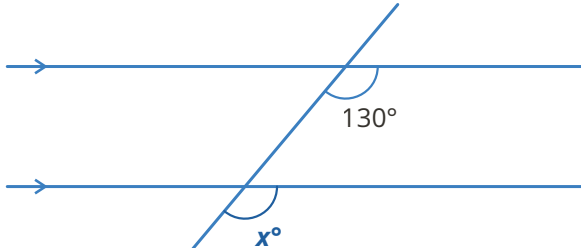
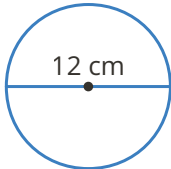
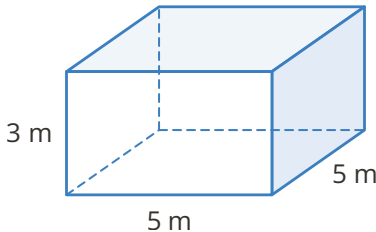
Week 9 · Session 3

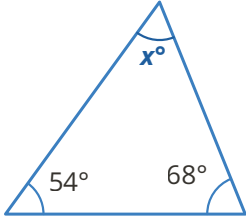
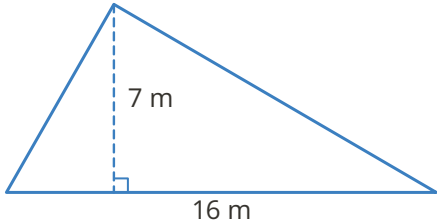
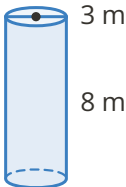
TERM 1

Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING						
1 Find the value of x. 	FORMULA Corresponding angles on parallel lines are equal COPY SUBSTITUTE ANSWER						
2 Find the circumference of this circle, correct to 1 decimal place. 	FORMULA $C = \pi d$ COPY SUBSTITUTE ANSWER						
3 Find the volume of this rectangular prism. 	FORMULA $V = lwh$ COPY SUBSTITUTE ANSWER						
4 Find the number. 15% of a number is 45	FORMULA Original amount = known amount $\div$ its percentage $\times 100$ COPY SUBSTITUTE ANSWER						
5 Find the mean of these scores. <table><tr><td>Scores</td><td>21</td><td>18</td><td>24</td><td>15</td><td>22</td></tr></table>	Scores	21	18	24	15	22	FORMULA Mean = $\frac{\text{sum of scores}}{\text{number of scores}}$ COPY SUBSTITUTE ANSWER
Scores	21	18	24	15	22		

QUESTION	WORKING
1 Find the value of x. 	FORMULA $a + b + c = 180^\circ$ COPY _____ SUBSTITUTE _____ ANSWER _____
2 Find the area of this triangle. 	FORMULA $A = \frac{1}{2}bh$ COPY _____ SUBSTITUTE _____ ANSWER _____
3 Find the volume of this cylinder, correct to 1 decimal place. 	FORMULA $V = \pi r^2 h$ COPY _____ SUBSTITUTE _____ ANSWER _____
4 Write as a single power. $(3^4)^2$	FORMULA $(a^m)^n = a^{mn}$ COPY _____ SUBSTITUTE _____ ANSWER _____
5 A fair die is rolled. Find the probability of rolling a multiple of 3. 	FORMULA $P(E) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$ COPY _____ SUBSTITUTE _____ ANSWER _____

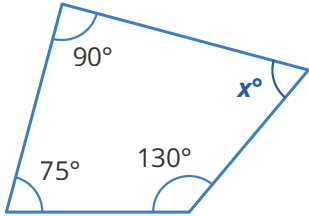
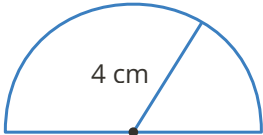
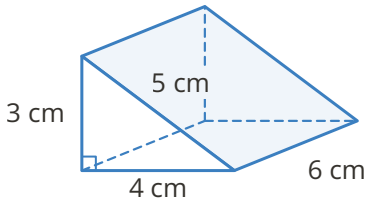
Week 10 · Session 2

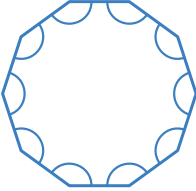
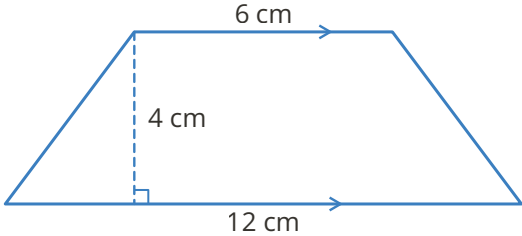
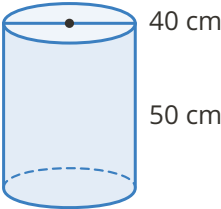
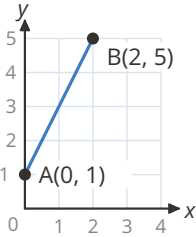
TERM 1

Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA $a + b + c + d = 360^\circ$ COPY _____ SUBSTITUTE _____ ANSWER _____
2 Find the area of this semicircle, correct to 1 decimal place. 	FORMULA $A = \frac{1}{2}\pi r^2$ COPY _____ SUBSTITUTE _____ ANSWER _____
3 Find the surface area of this triangular prism. 	FORMULA $SA = 2 \times \text{area of triangle} + \text{area of the 3 rectangles}$ COPY _____ SUBSTITUTE _____ ANSWER _____
4 Calculate. 5% of 180 m	FORMULA $\text{Amount} = \frac{\text{percentage}}{100} \times \text{quantity}$ COPY _____ SUBSTITUTE _____ ANSWER _____
5 Find the probability that it does not snow. $P(\text{snow}) = 0.08$	FORMULA $P(\text{not } E) = 1 - P(E)$ COPY _____ SUBSTITUTE _____ ANSWER _____

QUESTION	WORKING
1 Find the sum of the interior angles of this decagon. 	FORMULA Angle sum = $180^\circ \times (n - 2)$ COPY _____ SUBSTITUTE _____ ANSWER _____
2 Find the area of this trapezium. 	FORMULA $A = \frac{1}{2}h(a + b)$ COPY _____ SUBSTITUTE _____ ANSWER _____
3 Find the capacity of this tank in litres, correct to 1 decimal place. 	FORMULA $V = \pi r^2 h$, $1000 \text{ cm}^3 = 1 \text{ L}$ COPY _____ SUBSTITUTE _____ ANSWER _____
4 Expand. $2(x + 7)$	FORMULA $a(b + c) = ab + ac$ COPY _____ SUBSTITUTE _____ ANSWER _____
5 Find the gradient of line AB. 	FORMULA $m = \frac{\text{rise}}{\text{run}}$ COPY _____ SUBSTITUTE _____ ANSWER _____